

Half-degree irreducibles over $\mathbb{F}_2$: a proof roadmap with one missing bridge

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Status. This is not a proof of the Kaser–Lemire conjecture. The red estimate in Figure 1 remains open; all subsequent claims are conditional on it.

Abstract

Kaser and Lemire observed that Barrett reduction over $\mathbb{F}_2$ is especially cheap when an irreducible polynomial f of degree n satisfies $\deg(f - x^n) \leq \lfloor n/2 \rfloor$, and conjectured existence for every n . We give a compact audit of an attempted proof. Reciprocal polynomials, logarithmic differentiation, and Möbius inversion reduce the problem to a signed ray-class trace. A characteristic-two inverse-energy theorem supplies the wrapped bilinear input, and exact Haar and order decompositions eliminate all coarse and low-order frequencies. What remains is one uniform high-Witt signed-discrepancy estimate. We state it exactly and show how it would close both parity endpoints. Computation independently verifies $n \leq 400$ but is not used as evidence for the missing estimate.

1 The proof chain at a glance

Put $\ell = \lceil n/2 \rceil - 1$. The following diagram distinguishes theorem from experiment and conjectural bridge.

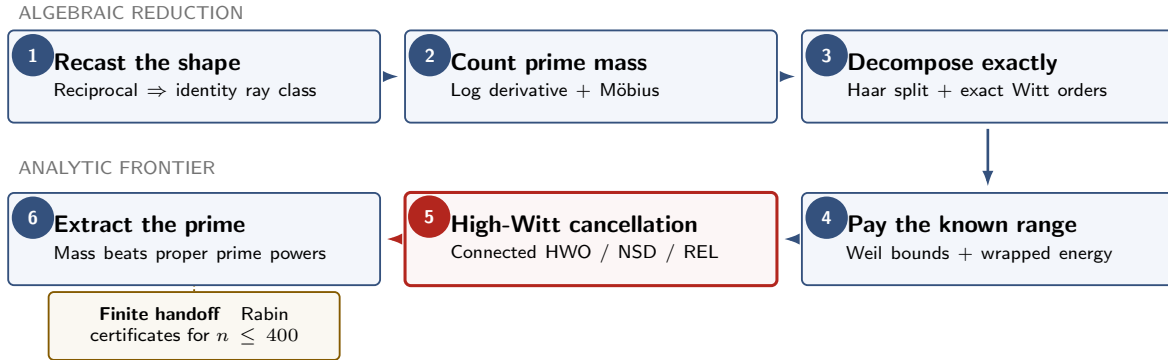


Figure 1: The current proof architecture. Stages 1–4 and 6 are proved; Stage 5 is the sole load-bearing open estimate. Gold denotes finite verification.

2 Proved reductions, step by step

1. **Reciprocal ray class (PROVED)**. For $j \geq 0$ set

$$\mathcal{E}_j = (1 + x\mathbb{F}_2[x])/(x^{j+1}), \quad |\mathcal{E}_j| = 2^j,$$

and, for monic F , let $\langle F \rangle_j = x^{\deg F} F(x^{-1}) \pmod{x^{j+1}}$. The bound $\deg(f - x^n) \leq \lfloor n/2 \rfloor$ is exactly $\langle f \rangle_\ell = 1$. Thus the conjecture asks for a degree- n prime in one prescribed Hayes ray class.

2. **Mangoldt population and prime extraction (PROVED)**. Define

$$N_j(b) = \sum_{\substack{F \text{ monic, } \deg F = n \\ \langle F \rangle_j = b}} \Lambda(F).$$

Exact logarithmic differentiation counts prime powers with weight Λ ; Möbius inversion, or direct subtraction here, then extracts primes. At the odd endpoint $n = 2\ell + 1$, $N_\ell(1) = 1 + nI_n(1)$. At the even endpoint $n = 2\ell + 2$, odd proper powers

vanish from the identity class and the remaining square/even-power mass is bounded explicitly. For $\ell \geq 200$, the sufficient common target is

$$N_\ell(1) > 2^{n-\ell} - 2^\ell. \quad (1)$$

3. Exact Haar reconstruction (PROVED). Each map $\mathcal{E}_j \rightarrow \mathcal{E}_{j-1}$ has two-element fibres. If $H_j(b)$ is the difference of the two child populations above b , repeated binary splitting gives

$$2^\ell N_\ell(e) = 2^n + \sum_{j=1}^{\ell} \varepsilon_j(e) 2^{j-1} H_j(b_j(e)), \quad \varepsilon_j(e) \in \{\pm 1\}. \quad (2)$$

For primitive characters X_j of conductor j ,

$$H_j(b) = 2^{1-j} \sum_{\chi \in X_j} \overline{\chi(b)} S_n(\chi), \quad |H_j(b)| \leq (j-1) 2^{\lceil n/2 \rceil},$$

the latter by the function-field Riemann hypothesis. This pays every coarse level, but taking absolute values at all levels loses a fatal factor ℓ .

4. Bilinear input (PROVED, but not the endpoint). The required characteristic-two inverse-additive energy is controlled even when polynomial products wrap modulo x^r . Collisions are stratified by x -adic valuation; Padé approximation converts each stratum to bounded-degree factorizations, after which lift and divisor estimates apply. This validates the Bagshaw-style Type-I/II ledger. It does *not* control the final signed Möbius convolution: pointwise use of the bilinear bound again discards the cancellation in (2).

5. Keep the top conductors connected (PROVED reduction). Let $c = \lceil \log_2 \ell \rceil$, $a = \ell - c - 1$, and

$$W_{\ell,n} = 2^{\lceil n/2 \rceil} \sum_{1 \leq j < a} (j-1) 2^{j-1}, \quad B_{\ell,n} = 2^{2\ell} - W_{\ell,n} > 0.$$

The identity path telescopes without absolute values:

$$C_{\ell,n} := \sum_{j=a}^{\ell} 2^{j-1} H_j(1) = 2^\ell N_\ell(1) - 2^{a-1} N_{a-1}(1). \quad (3)$$

Hence the one-sided estimate

$$C_{\ell,n} > -B_{\ell,n} \quad (\text{REL})$$

implies (1): the paid lower levels contribute at worst $-W$, so $2^\ell N_\ell(1) - 2^n > -W - B = -2^{2\ell}$.

6. Resolve exact Witt orders (PROVED reduction). For $s \geq 0$, write

$$\mathcal{H}_{j,s} = \{\chi : \chi^{2^s} = 1\}, \quad h_{j,s} = 2^{j-\lfloor j/2^s \rfloor}, \quad P_{j,s} = \sum_{g \in 2^s \mathcal{E}_j} N_j(g).$$

Finite-group orthogonality identifies the exact conductor- j , exact order- 2^s trace as

$$T_{j,s} = h_{j,s} P_{j,s} - h_{j,s-1} P_{j,s-1} - h_{j-1,s} P_{j-1,s} + h_{j-1,s-1} P_{j-1,s-1}.$$

Let Q be the largest power of two with $3Q \lceil \log_2 \ell \rceil \leq \ell$. Individual Weil bounds pay every layer of order at most Q . Only $O(\log \log \ell)$ genuinely high-Witt order bands per top conductor survive. At $\ell = 200$, 20 of 67 layers are discharged and 47 remain.

3 The missing bridge

7. High-Witt order-layer estimate (OPEN). One sufficient uniform statement, for $a \leq j \leq \ell$, $2^s > Q$, and every nonempty exact layer, is

$$4\ell |T_{j,s}(n)| \leq \#X_{j,s}(j-1) 2^{\lceil n/2 \rceil}. \quad (\text{HWO})$$

There is an equivalent sparse-discrepancy form. Put $\Delta_{j,s} = 2P_{j,s} - P_{j-1,s}$, $d_s = \lfloor (j-1)/2^s \rfloor$, and $R_{j,s} = 2^{d_{s-1}-d_s}$. Exact layer counting gives

$$4\ell |R_{j,s} \Delta_{j,s} - \Delta_{j,s-1}| \leq (R_{j,s} - 1)(j-1) 2^{\lceil n/2 \rceil}. \quad (\text{NSD})$$

For surviving orders, $2^s > \ell/(3c)$, so $2^s \mathcal{E}_j$ has fewer than $8\ell^3$ elements. This is a sharp localization, not an averaging proof: at the largest exact order, the subgroup is $\{1\}$ and NSD still contains the original identity-ray increment.

The implication chain is now formal and short:

$$\boxed{\text{HWO (or another proof of REL)}} \implies \text{REL} \implies (1) \implies \text{prime in the identity class.}$$

The last implication uses the proper-power bounds in Step 2. Degrees $n \leq 400$ are separately covered by checked Rabin irreducibility certificates (**FINITE**). Therefore HWO, uniformly for $\ell \geq 200$ and $n \in \{2\ell + 1, 2\ell + 2\}$, would prove the conjecture in every degree.

What has been ruled out. Separate characterwise or conductorwise absolute values are too expensive; support-only L^4 bounds are exactly trivial; naive Cauchy, bent/Kerdock, simple Heisenberg lifts, and several stronger finite diagnostics fail. Existing large-field moment theorems do not supply a fixed-field, growing-conductor, higher-Witt estimate uniform in ℓ . The credible target is therefore a connected characteristic-two trace or cohomological object that proves (HWO), (NSD), or (REL) directly. More finite tables alone do not advance that theorem.

References

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